WGS/WES best practices

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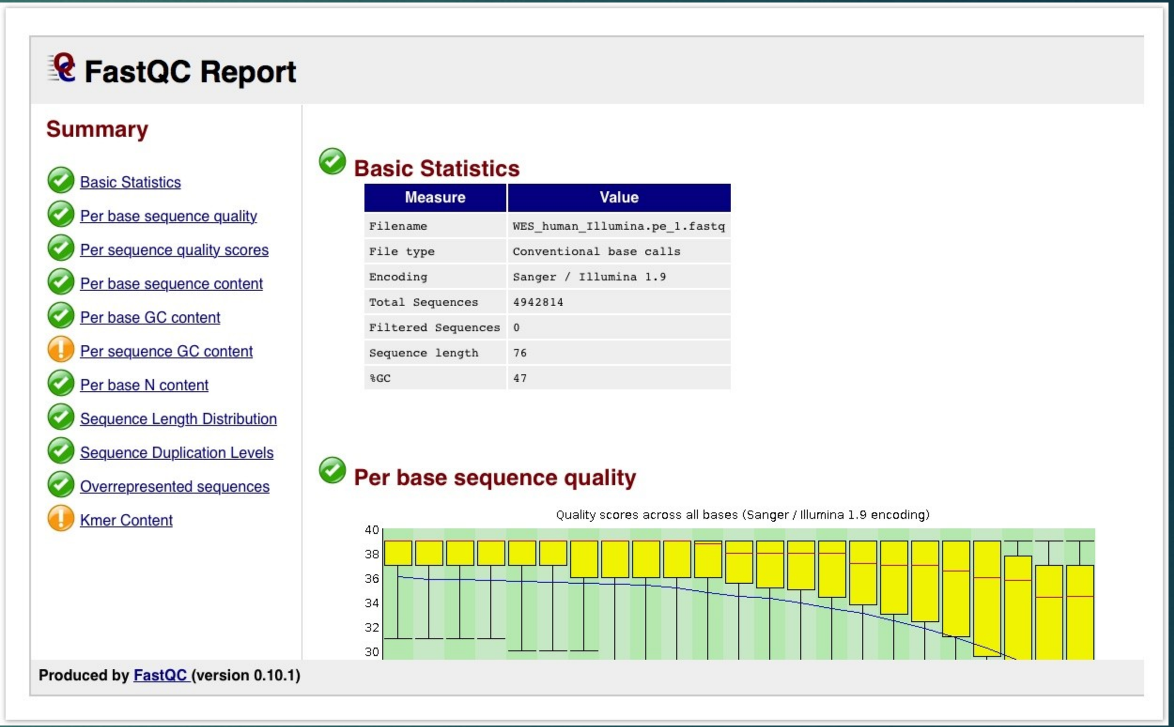
Experimental design

- Coverage for germline mutations
  - WES: 100X coverage, where on average, each base in the exonic region can be read 100 times
  - WGS: 30X coverage
- Coverage for somatic (tumor/normal)
  - WES: 100X-150X
  - WGS: 100X for tumor samples
- If looking at somatic mutations, must have tumor and normal DNA from same patient.
- Sample size depends heavily on study objectives, effect sizes, and statistical power
  - Common variants require fewer samples.
  - Strong effect (eg. Mendelian Disease) requires fewer samples

Best practices during quality control

- **Purpose:** Assess the quality of the reads and ensure that the sequencing data is of high quality and reliable for downstream analysis.
- **FastQC**: Identifies potential issues such as poor read quality, adapter contamination, and sequence biases.
- **Fastq_screen**: Aligns reads to multiple reference genomes/databases. Good way to detect contamination.
- **Adapter trimming**: Adapter trimming is optional, but often recommended. CutAdapt is the most common tool used for adapter trimming.



WES vs. WGS

- WES focuses on the coding regions of the genome.
  - Often used in smaller studies where cost is a concern.
  - Difficult time capturing structural variants, mutations in non-coding region.
  - Used as first-line, quicker analysis compared to WGS
- WGS sequences the entire genome
  - Used when cost and time aren't a problem
  - Captures structural variants and mutations in non-coding regions

Best practices during alignment

- BWA (BWA-MEM2) – Most popular for WGS/WES
  - Pros: Accurate, works well with GATK
  - Cons: Can be slow and computationally intensive
- Bowtie2 – High speed and memory efficient aligner.
  - Pros: Fast, not as computationally intensive
  - Lower sensitivity than BWA-MEM
- Novoalign – Closed source, high accuracy
  - Pros: Very accurate and reliable
  - Cons: Can be slow and requires a license
- DRAGEN – Closed source, high accuracy, fast
  - Pros: Accuracy, speed, alignment and variant/copy number calling all in one step
  - Cons: Requires a license

Studies comparing aligners

- Betschart et al – Benchmark study comparing BWA-MEM with DRAGEN
  - Study showed that alignment and mappings step was more accurate and faster using DRAGEN vs. BWA-MEM2/GATK.
- Barbitoff et al – Benchmark study comparing various alignment tools and variant callers
  - Variant callers have a much larger effect on variant than choice of aligner.
  - Mentioned that BWA-MEM2 is a reliable aligner
  - Bowtie2 shouldn't be used for human variant calling, significantly worse performance than other aligners.

Post-alignment processing

- Sorting and indexing bam files for downstream analysis
- Marking duplicates
  - Read duplicates give false confident in a specific variant and can affect copy number calling
- Base quality score recalibration
  - Sequencing techniques have certain biases for determining quality scores. Tools like GATK BaseRecalibrator help correct for this

Best practices during variant calling

- GATK (HaplotypeCaller) – Most popular tool for germline variant calling
  - Pros: Good sensitivity for SNVs and indels, BQSR included
  - Computationally very slow, requires multiple steps
- FreeBayes – Germline variant caller sed for small datasets
  - Pros: Can call complex variants, works well on non-model organisms
  - Slower than GATK, high false positive rate
- Strelka2 – Fast, mainly used for somatic variants
  - Pros: Fast and memory-efficient, high accuracy
  - Not well suited for structural variant detection

Best practices during variant calling

- Mutect2 (GATK) – Somatic variant calling in cancer
  - Pros: Good sensitivity for low-frequency SNVs and indels. Works well with matched normal-tumor samples
  - Computationally very slow and intensive, prone to false positives
- DeepVariant (Google) – Germline variant caller
  - Pros: Highly accurate due to deep learning approach, fewer false positives, open-source
  - Cons: Requires GPU acceleration for best performance
- DRAGEN (Illumina)
  - Pros: Very fast, integrated QC and filtering, alignment and variant calling all in one step
  - Cons: Requires license and expensive hardware dependency

Studies comparing variant callers

- Betschart et al - Benchmark study comparing GATK/BWA-MEM, DeepVariant, and DRAGEN for germline variant calling
  - GATK had the lowest performance of the three
  - However, DeepVariant requires GPU acceleration and DRAGEN requires a license
- Barbitoff et al – Benchmark study comparing several germline variant callers
  - Modern machine learning models like DeepVariant outperform traditional models like GATK, even when using GATK’s machine learning model

Studies comparing variant callers

- Abdelwehab et al – Compared 5 germline variant callers for short and long reads
  - Short reads - DeepVariant provided the most balanced variant calling performance, GPU compatible. Highest F1 score for SNVs and INDELs
  - GATK4 was the most conventional tool, allows for manual filtering and thresholding
  - Long reads – DeepVariant outperformed conventional methods, especially for indels. DNAscope is a computationally efficient AI tool that also performed well.

Studies comparing variant callers

- Guille et al – Compared 20 somatic variant callers for WES datasets with reference mutations
  - Reads were aligned using BWA for all variant callers and base qualities were recalibrated using GATK
  - 20 tools were used including widely used tools, commercial tools, deep learning tools
  - Top performers for somatic SNVs: Dragen, Muse, TNScope, Mutect2
  - Top performers for somatic indels: NeuSomatic, DeepSomatic, DRAGEN
  - DRAGEN was the most robust and also was one of the fastest
  - Commercial tools generally outperformed open-source tools

Germline vs Somatic variants

- Different Assumptions: Germline variant callers (e.g., GATK HaplotypeCaller, DeepVariant) assume diploidy with allele frequencies of ~50% or 100%, while somatic callers (e.g., Mutect2, Strelka2) detect low-VAF mutations and require stricter filtering.
- Matched Normal Requirement: Somatic callers often need a matched normal sample to filter out inherited variants, whereas germline callers use population databases for filtering.
- Error Handling & Sensitivity: Somatic callers are optimized for low-frequency mutations and tumor heterogeneity, while germline callers focus on high-confidence inherited variants

Structural variants/Copy number variantion

- Gabrielaite et al. – Study comparing 11 tools for copy-number variation detection in germline WGS and WES data.
- GATK gCNV had high recall but low precision.
- Lumpy and DELLY had more balanced recall and precision, reducing false positives.
- Manta and CNVkit performance varied a lot across datasets
- Study suggested that using multiple tools is the gold standard

Variant Filtering

- Pedersen et al. - Study evaluates variant filtering strategies using two WGS and WES family cohorts.
- Genotyping Quality (GQ >= 20) – ensures confidence in a genotype call
- Allele Balance (AB: 0.2-0.8) – Extreme values could suggest sequencing bias or error
- Sequencing Depth (DP >= 10) - Minimum sequencing depth covering a variant site to reduce false positives
- Population Allele Frequency (<0.001 for dominant and <0.01 for recessive)

Annotation

- **Functional Annotation:** Predicts the biological impact of variants using tools like SIFT, PolyPhen, and CADD to assess protein function and regulatory effects.
- **Population & Clinical Databases:** Filters variants based on population allele frequencies from gnomAD and 1000 Genomes and links to disease databases like ClinVar and COSMIC.
- **Pathogenicity:** Uses quality filters and pathogenicity scoring to identify rare, high-impact variants relevant to disease or functional studies.

- GATK –genotype gVCF can genotype across samples.
- Long read alignment, minimap